

Agenda

1. Friday Feedback →
2. Section 9.5
3. Section 11.1



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Recall

linear diff eq.

$$\frac{dy}{dx} + P(x)y = Q(x)$$

Integrating factor $I(x) = e^{\int P(x) dx}$ ↖ no + C

Activity: $t \frac{dy}{dt} + 4y = t^5$, for $t > 0$

$$\frac{dy}{dt} + \frac{4}{t}y = t^4$$

$$P(t) = \frac{4}{t} \Rightarrow \int P(t) dt = 4 \ln t$$

$$I(t) = e^{4 \ln t} = e^{\ln t^4} = t^4$$

(log property)

① Multiply by $I(t)$

$$\left(\frac{dy}{dt} + \frac{4}{t}y = t^4 \right) t^4 \rightarrow t^4 \frac{dy}{dt} + 4t^3 y = t^8$$

② product rule + Integrate

$$\int \frac{d}{dx} [t^4 y] = \int t^8$$

$$t^4 y = \frac{1}{9} t^9 + C$$

④ solve

$$y = \frac{\frac{1}{9} t^9 + C}{t^4} = \boxed{\frac{1}{9} t^5 + \frac{C}{t^4}}$$

Example: $\frac{dv}{dt} - 2t v = 3t^2 e^{t^2}$ $V(0) = 6$

want: $\frac{dv}{dt} + P(t)v = Q(t)$

$$P(t) = -2t \rightarrow \int P(t) dt$$

$$= \int -2t dt$$

$$= -t^2$$

$$I(t) = e^{-t^2}$$

$$\frac{e^{-t^2}}{1} \frac{dv}{dt} - \frac{2t e^{-t^2}}{1} \frac{v}{2} = 3t^2 \frac{e^{-t^2}}{e^{-t^2}}$$

$$\int \frac{d}{dt} [e^{-t^2} v] = \int 3t^2$$

$$e^{-t^2} v = t^3 + C$$

initial condition $v(0)$

$$e^{-(0)^2} (6) = (0)^3 + C$$

$$6 = C$$

$$e^{-t^2} v = t^3 + 6$$

$$v = (t^3 + 6) e^{t^2}$$

Section 11.1 - Sequences 😊

Examples:

1.) $\{1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots\}$

$n=1 \quad n=2 \quad n=3 \quad n=4$

$a_n = \frac{1}{n}, \quad n \geq 1, n \in \mathbb{Z}$

Sometimes brackets,
sometimes don't - inconsistent.

2.) $2, 4, 6, 8, \dots$

$a_n = 2n, \quad n \geq 1, n \in \mathbb{Z}$

Definition:

An infinite sequence of Real numbers
if a function whose domain is the set
of positive integers.

It is a function $a(n)$ for $n \geq 1$ where
 $a(1), a(2), a(3), \dots$ becomes $a_1, a_2, a_3, \dots$

indices

first three

Activity: Write the ~~three~~ ~~four~~ terms of each sequence

1.) $\{a_n\} = \left\{ \frac{1}{3n-2} \right\} \quad \frac{1}{3(1)-2}, \frac{1}{3(2)-2}, \frac{1}{3(3)-2} \rightarrow a_1, \frac{1}{4}, \frac{1}{7}$

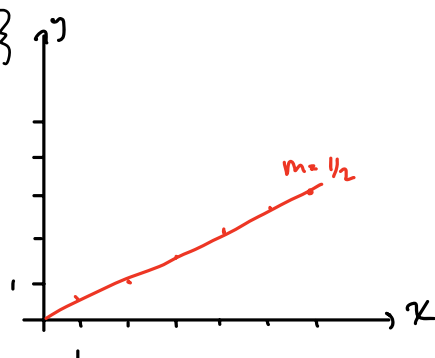
2.) $\{b_n\} = \left\{ \frac{2n-1}{n^3} \right\} \quad \frac{2(1)-1}{1^3}, \frac{2(2)-1}{2^3}, \frac{2(3)-1}{3^3} \rightarrow b_1, \frac{3}{8}, \frac{5}{27}$

3.) $\{c_n\}_{n=0}^{\infty} = \left\{ (-1)^n \left(\frac{1}{2} \right)^n \right\}_{n=0}^{\infty} \rightarrow c_0, c_1, c_2$

Graphs of Sequences

Examples: Graph the sequences both in the xy-plane
and on the number line

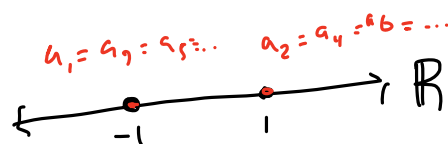
1.) $\{a_n = \frac{n}{2}\}$



diverges $\rightarrow \infty$

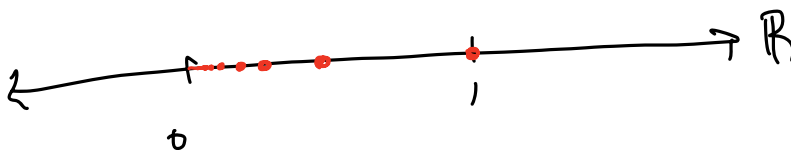
2.) $\{a_n = (-1)^n\}$

$\{-1, 1, -1, 1, -1, \dots\}$



diverges

3.) $\{a_n = \frac{1}{n}\}$



Converges to 0

Convergence Let $\{a_n\}$ be a sequence

1.) We say that $\{a_n\}$ converges to L , written as $\lim_{n \rightarrow \infty} a_n = L$ if, for every $\varepsilon > 0$, there is a positive integer N such that $|a_n - L| < \varepsilon$ for all $n \geq N$

2.) We say that $\{a_n\}$ diverges if $\lim_{n \rightarrow \infty} a_n$ does not exist

Theorem: If $f(x)$ is a real-valued function where $f(n) = a_n$ for all positive integers n and $\lim_{x \rightarrow \infty} f(x) = L$ then $\lim_{n \rightarrow \infty} f(n) = L$

Activity: $\lim_{n \rightarrow \infty} \frac{3n^3 + 7n^2 + 1}{4n^3 + 8n + 8}$

Example: $\lim_{n \rightarrow \infty} \cos(2\pi n)$